

AC2205 Marking Scheme

Intermediate Management Accounting · Summer 2024 · Paper 1

Examination Session and Year	Summer 2024
Module Code	AC2205
Module Title	Intermediate Management Accounting
Paper Number	1
External Examiner	Professor Martin Quinn
Head of School/Department	Professor Michelle Carr
Internal Examiners	Mr. Conor Loughnane
Duration of Paper	1.5 hours
Special Requirements	Students are allowed to use calculators that have standard arithmetic, scientific, graphical or algebraic functions provided that they are NOT programmable and can NOT store text (other than algebraic notation).

Instructions to Candidates (as per exam paper): Answer two questions only. All questions carry equal marks (50 each). Show all your workings.

Own-figure rule: where a candidate makes an error at an early stage of a multi-step numerical question, full method marks are awarded for correctly applying the required technique to their own (incorrect) figure in all subsequent steps. Marks are never cascaded away for a single early error.

Marking philosophy: this is a closed-book, time-pressured examination. Academic citations are never required. Discursive answers are marked on depth of explanation and application to the specific scenario, not generic textbook recall.

Question Overview

Question	Scenario	Topics	Marks
Question 1	BlueSky Ltd	Cash budgeting with a minimum balance requirement and short-term borrowing/repayment mechanics	50
Question 2	Baldwin Shoe Company	Relevant costing, segment discontinuation decision, avoidable vs unavoidable costs, opportunity cost	50
Question 3	SmartWearables Ltd	Flexible budgeting, flexible budget and volume variances, performance evaluation, disadvantages of budgeting	50

Candidates answer any TWO of the three questions above; each is worth 50 marks. This marking scheme covers all three questions in full.

QUESTION 1

BlueSky Ltd

Total: 50 marks

BlueSky Ltd manufactures high-tech drones. Candidates are required to (a) prepare a cash budget for Quarter Four 2024, including a minimum monthly cash balance requirement met through short-term borrowing (with interest charged on repayment), and (b) analyse the resulting cash position with advice to management.

Part	Requirement	Marks
(a)	Cash budget for Q4 (Oct, Nov, Dec), incorporating the €22,000 minimum balance and borrowing/repayment mechanics	40
(b)	Analysis of the Q4 cash position, with advice to management	10

Question 1(a)**40 marks**

BlueSky Ltd, budgeted figures July–December (€): Sales, Purchases, Salaries, Marketing, Administration and Overheads given monthly. Credit sales (60% of sales) settled 60% in month of sale, 20% one month after, 20% two months after; cash sales (40%) receive a 3% discount. Credit purchases (50% of purchases) paid 70% in month of purchase, 10% one month after, 20% two months after. Salaries paid one month in arrears; marketing paid two months in arrears; overheads (including €7,000/month depreciation) paid one month in arrears; administration paid as incurred. €45,000 investment income expected in October. New equipment purchased June 2024 for €350,000, paid in 10 equal monthly instalments from July 2024. €28,000 tax payment due in October. Old equipment sold for €25,000 in November and €50,000 in December. Cash balance at 30 September 2024 expected to be €20,000. From 1 October 2024, a minimum monthly closing cash balance of €22,000 applies, met by short-term borrowing (in €1,000 multiples) on the first day of the month, repaid on the last day of a later month, with 12% p.a. interest charged on repayment for the period outstanding. (a) Prepare a cash budget for Quarter Four 2024.

Full Model Answer - Working 1: cash receipts from sales**Cash receipts from sales**

Item	October (€)	November (€)	December (€)
Cash sales this month, net of 3% discount (40% of month's sales × 97%)	124,160	135,800	147,440
Credit sales settled in month of sale (60% of month's sales × 60%)	115,200	126,000	136,800
Credit sales settled one month after (prior month's credit sales × 20%)	36,000	38,400	42,000
Credit sales settled two months after (two months prior credit sales × 20%)	33,600	36,000	38,400
Total cash receipts from sales	308,960	336,200	364,640

Working detail (October shown): October cash sales: $40\% \times €320,000 = €128,000$, less 3% discount = €124,160. October credit sales settled this month: $60\% \times €320,000 \times 60\% = €115,200$. Settled one month after (from September's credit sales): $60\% \times €300,000 \times 20\% = €36,000$. Settled two months after (from August's credit sales): $60\% \times €280,000 \times 20\% = €33,600$.

Full Model Answer (continued) – Working 2: cash payments**Cash payments for purchases**

Item	October (€)	November (€)	December (€)
Cash purchases this month (50% of month's purchases, paid same month)	90,000	100,000	110,000
Credit purchases paid same month (50% of month's purchases × 70%)	63,000	70,000	77,000
Credit purchases paid one month after (× 10%)	8,000	9,000	10,000
Credit purchases paid two months after (× 20%)	14,000	16,000	18,000
Total cash payments for purchases	175,000	195,000	215,000

Other payments: salaries, marketing, overheads, administration

Item	October (€)	November (€)	December (€)
Salaries (one month in arrears: Sep, Oct, Nov paid)	36,000	38,000	40,000
Marketing (two months in arrears: Aug, Sep, Oct paid)	28,000	31,000	33,000
Overheads, excl. €7,000 depreciation, one month in arrears (Sep, Oct, Nov generated)	27,000	29,000	31,000
Administration (paid as incurred, same month)	24,000	26,000	28,000

Overheads note: As with the depreciation adjustment seen in similar questions, the €7,000 monthly depreciation embedded in the overheads figure is a non-cash charge and is stripped out before applying the one-month payment lag: September overheads generate a cash payment of €34,000 – €7,000 = €27,000, paid in October, and so on.

Other cash items

Item	October (€)	November (€)	December (€)
Equipment instalment (€350,000 ÷ 10, months 4, 5 and 6 of the 10-month schedule beginning July)	35,000	35,000	35,000
Tax payment	28,000	0	0
Investment income received (cash inflow)	45,000	0	0
Sale of old equipment (cash inflow)	0	25,000	0
Sale of old machine (cash inflow)	0	0	50,000

Equipment instalment timing: The equipment was purchased in June 2024 with 10 equal monthly instalments beginning in July 2024 — so instalments run from July through to April 2025. October, November and December fall within this window (the 4th, 5th and 6th instalments respectively), so all three months carry the full €35,000 payment.

Full Model Answer (continued) – Cash position before financing, and the borrowing/repayment schedule

Cash budget before financing activity

Item	October (€)	November (€)	December (€)
Total receipts	353,960	361,200	414,640
Total payments	353,000	354,000	382,000
Net cash flow (before financing)	960	7,200	32,640
Opening cash balance	20,000	22,960	28,120
Balance before financing	20,960	30,160	60,760

October – shortfall requires borrowing

The pre-financing balance of €20,960 is below the €22,000 minimum by €1,040. Since funds can only be borrowed in €1,000 multiples, borrowing exactly €1,000 would leave a balance of €21,960 — still below the minimum — so BlueSky Ltd must borrow **€2,000**, the smallest multiple of €1,000 sufficient to reach the €22,000 minimum. This gives a closing October balance of €20,960 + €2,000 = **€22,960**. No interest is charged in the month of borrowing itself, since interest is only calculated when the principal is repaid.

November – surplus allows full repayment

The pre-financing balance of €30,160 is €8,160 above the €22,000 minimum, comfortably enough to repay the entire €2,000 loan outstanding from October without breaching the minimum balance. The loan was outstanding from 1 October to 30 November — two full months — so interest = $€2,000 \times 12\% \times (2/12) = \textbf{€40}$. Closing November balance = €30,160 – €2,000 (principal) – €40 (interest) = **€28,120**.

December – no financing activity required

With the October loan fully repaid, there is no outstanding balance to service, and the pre-financing balance of €60,760 is well above the €22,000 minimum, so no further borrowing or repayment is needed. Closing December balance = **€60,760**.

Cash budget for Quarter Four 2024, including financing activity

Item	October (€)	November (€)	December (€)	Quarter total (€)
Balance before financing	20,960	30,160	60,760	
Add: Amount borrowed	2,000	0	0	2,000
Less: Amount repaid	0	(2,000)	0	(2,000)
Less: Interest paid on repayment	0	(40)	0	(40)
Closing cash balance	22,960	28,120	60,760	60,760

Quarter as a whole: Starting from the opening balance at 1 October (€20,000), the net operating cash flow across the quarter (€960 + €7,200 + €32,640 = €40,800) plus the net effect of borrowing and repaying the short-term loan (€2,000 borrowed less €2,000 repaid less €40 interest = €(40) net) gives €20,000 + €40,800 – €40 = €60,760, matching the December closing balance exactly.

Mark Allocation

Marks	Criteria
34-40	All receipts and payments correctly calculated and timed (including the depreciation adjustment and equipment instalment window); minimum balance requirement correctly identified as breached in October; borrowing correctly rounded up to €2,000; repayment and 2-month interest calculation in November correct; all monthly and quarterly closing balances correct.
25-33	Substantially correct with minor timing or computational errors, most commonly in the borrowing/repayment rounding or the interest period calculation.
13-24	Core cash budget structure correct but the minimum balance/borrowing mechanic misunderstood or omitted (e.g. borrowing not rounded to €1,000 multiples, or interest not calculated on the correct outstanding period).
0-12	Limited understanding of cash budgeting; minimal correct workings.

Question 1(b)**10 marks**

Based upon your answer to (a) above, briefly analyse BlueSky Ltd.'s budgeted Quarter Four cash position, offering whatever advice to management that you consider appropriate.

Full Model Answer**Overall assessment**

BlueSky Ltd's underlying cash position improves steadily and substantially across the quarter: a marginal October shortfall requiring a small €2,000 short-term loan gives way to a healthy and growing surplus by November and December, ending the quarter with a closing balance of €60,760 — almost three times the required €22,000 minimum. The need to borrow at all was very short-lived (a single month) and modest in scale.

What is driving the improvement

Sales are budgeted to grow steadily through Q4 (from €320,000 in October to €380,000 in December), which increasingly outpaces the growth in payments, particularly once the one-off €28,000 tax payment (October only) drops out of the picture and the €45,000 investment income and equipment sale proceeds (€25,000 in November, €50,000 in December) provide additional one-off boosts to cash in the later months.

Advice to management

1. The modest, short-lived October borrowing requirement is well managed by the existing minimum-balance policy and does not, on this budget, indicate a deeper cash management problem — management can be reasonably confident in the plan as it stands. **2.** Given the improving trend, management should consider whether the €22,000 minimum balance policy remains appropriate for November and December, when the business is running well ahead of it, or whether surplus cash could be more productively used (e.g. early repayment of other liabilities, or short-term investment) once the loan is cleared. **3.** As with any budget, actual results should be closely monitored against this plan, particularly the sales growth assumption driving much of the improvement, since a shortfall in actual sales relative to budget could quickly re-introduce the need for further short-term borrowing.

Mark Allocation

Marks	Criteria
9-10	Correctly identifies the improving trend and the modest, temporary nature of the October borrowing; explains the key drivers (sales growth, one-off receipts); gives specific, sensible advice referencing the figures in part (a).
6-8	Correctly identifies the overall position with reasonable explanation of drivers; advice more generic.
3-5	Some relevant commentary but lacks specific reference to the figures produced in part (a).
0-2	Minimal or no meaningful commentary.

QUESTION 1 TOTAL — 50 MARKS

QUESTION 2**Baldwin Shoe Company**

Total: 50 marks

Baldwin Shoe Company operates men's, women's and children's shoe departments from a single store. The children's department has operated at an accounting loss and management is considering discontinuing it. Candidates are required to (a) calculate the children's department's contribution using relevant costs and recommend a decision, (b) confirm that decision via a full income statement comparison, (c) assess how an alternative use of the freed-up space affects the decision, and (d) outline the three criteria for a relevant cost.

Part	Requirement	Marks
(a)	Children's department relevant-cost contribution; keep or discontinue decision	10
(b)	Company income statement without the children's department, compared to the current position	15
(c)	Effect of an alternative use for the freed-up space on the decision	10
(d)	Three criteria for a relevant cost/benefit	15

Question 2(a)**10 marks**

Baldwin Shoe Company: Men's, Women's and Children's department income statements given (sales, COGS, gross margin, department manager's salary, sales commissions, rent on store lease, store utilities, net income/loss). Rent and utilities are shared equally across all departments; sales commissions and manager salaries are specific to each department. Children's department net loss: €(12,775). (a) Calculate the children's department's contribution to profit using relevant costs. Determine whether to keep operating or discontinue the children's department.

Full Model Answer**Identifying avoidable (relevant) vs unavoidable (irrelevant) costs**

Cost of goods sold, the department manager's salary, and sales commissions are all specific to the children's department — each department has its own dedicated manager and sales staff, and commissions are described as 'individual to each department.' These costs would genuinely disappear if the children's department closed, so they are **avoidable** and relevant to the decision. Rent on the store lease and store utilities, by contrast, are described as 'shared equally across all departments' — the total store rent and utilities bill does not change based on how many departments operate within it. These are **unavoidable (common) costs** that would continue regardless of the decision, and are therefore irrelevant to it.

Working - children's department relevant-cost contribution

Item	€
Sales revenue (lost if discontinued)	170,000
Less avoidable costs:	
Cost of goods sold	(98,375)
Department manager's salary	(24,000)
Sales commissions	(29,400)
Contribution to profit (relevant-cost basis)	18,225

DISAGREE

The children's department should be **kept operating**, not discontinued. Although the department shows an accounting net loss of €(12,775) once the full €31,000 of allocated rent and utilities is deducted, its relevant-cost contribution is a **positive €18,225**. This means the department currently earns enough to more than cover its own avoidable costs and make a positive contribution towards the store's unavoidable common costs (rent and utilities) and overall profit — discontinuing it would make Baldwin Shoe Company worse off by €18,225, since the €31,000 of rent and utilities would still have to be paid regardless.

Mark Allocation

Marks	Criteria
9-10	Avoidable costs correctly identified (COGS, manager salary, commissions) and unavoidable costs correctly excluded (rent, utilities); relevant contribution of €18,225 correctly calculated; correct recommendation (keep) with clear reasoning distinguishing the relevant-cost view from the accounting loss.
6-8	Substantially correct with a minor error, e.g. one cost item misclassified.
3-5	Some understanding of avoidable vs unavoidable costs shown, but the calculation or recommendation is incorrect or incomplete.
0-2	Limited understanding of relevant costing; minimal correct workings.

Question 2(b)**15 marks**

Confirm the conclusion you reached in part (a) by preparing an income statement for the company without the children's department and compare with the income statement in the information provided above for the Men's and Women's Department.

Full Model Answer

Key point on rent and utilities: Because rent and utilities are shared, company-wide costs that do not reduce if the children's department closes, the *full* company rent ($€24,000 \times 3 = €72,000$) and utilities ($€7,000 \times 3 = €21,000$) must still be charged in full against the remaining two departments' combined results — they are not simply dropped from the income statement.

Company income statement WITHOUT the children's department

Item	€
Sales revenue (Men's €630,000 + Women's €450,000)	1,080,000
Cost of goods sold ($€267,000 + €177,600$)	(444,600)
Gross margin	635,400
Department managers' salaries ($€55,000 + €44,000$)	(99,000)
Sales commissions ($€109,200 + €78,600$)	(187,800)
Rent on store lease (full company total, unchanged)	(72,000)
Store utilities (full company total, unchanged)	(21,000)
Net income - company without children's department	255,600

Comparison - net income WITH the children's department (current position)

Item	€
Men's Department net income	167,800
Women's Department net income	118,800
Children's Department net loss	(12,775)
Total company net income - current (with children's department)	273,825

Confirms part (a): Total company net income is €273,825 *with* the children's department, versus €255,600 *without* it — a difference of €18,225 in favour of keeping the department, exactly matching the relevant-cost contribution calculated in part (a). This confirms that the children's department should be kept operating: despite its reported accounting loss, discontinuing it would reduce, not improve, overall company profit.

Mark Allocation

Marks	Criteria
13-15	Full, correctly structured income statement without the children's department; rent and utilities correctly retained at their full, unreduced company-wide total; correct net income of €255,600; correct comparison to the current €273,825 total, correctly confirming the €18,225 difference matches part (a).
9-12	Substantially correct with a minor error, most commonly incorrectly reducing rent/utilities in the 'without' scenario.
5-8	Some correct workings but rent/utilities treatment fundamentally misunderstood, or comparison to part (a) not clearly drawn.
0-4	Limited understanding; minimal correct workings.

Question 2(c)

10 marks

Eliminating the children's department would increase space available to display men's and women's boots. Suppose management estimates that a wider selection of adult boots would increase the store's net earnings by €35,000. Would this information affect the decision that you made in part (a)?

Full Model Answer

The freed-up space represents an opportunity benefit

The €35,000 of additional net earnings from expanding the adult boot selection is only available *if* the children's department is discontinued and its space freed up — it is therefore a relevant benefit of the discontinuation decision, not a cost. It must be weighed directly against the €18,225 relevant contribution that would be lost by closing the children's department (from part (a)).

Working - revised decision including the opportunity to expand adult boots

Item	€
Additional net earnings from expanded adult boot space (benefit of discontinuing)	35,000
Less: Relevant contribution forgone by discontinuing children's department (from part (a))	(18,225)
Net benefit of discontinuing children's department and expanding adult boots	16,775

AGREE

Yes, this new information changes the decision. With the alternative use of the space now available, discontinuing the children's department and reallocating the space to men's and women's boots produces a **net benefit of €16,775** to the company — the €35,000 gain from the expanded adult boot selection more than offsets the €18,225 contribution given up by closing the children's department. On this new information, Baldwin Shoe Company should now discontinue the children's department.

Mark Allocation

Marks	Criteria
9-10	Correctly recognises the €35,000 as an opportunity benefit of discontinuing, weighs it directly against the €18,225 contribution from part (a), correctly calculates the €16,775 net benefit, and correctly concludes the decision now reverses to discontinue.
6-8	Correct recognition that the decision changes with a minor computational error.
3-5	Some relevant reasoning but the comparison to part (a)'s figure not clearly made, or the conclusion not clearly stated.
0-2	No meaningful engagement with how the new information affects the decision.

Question 2(d)**15 marks**

Outline and briefly explain the THREE criteria that must be met for any cost (or benefit) to be considered as relevant for decision-making purposes.

Full Model Answer**1. It must be a future cost or benefit**

A relevant cost or benefit must relate to something that will occur in the future, as a direct consequence of the decision being made now. Costs already incurred in the past (sunk costs) cannot be changed by any decision made today and are therefore never relevant — this is precisely why the children's department's historical cost of goods sold and past salary payments are irrelevant in themselves; what matters is whether these costs will continue to be incurred *going forward* if the department stays open.

2. It must differ between the alternatives being considered (it must be incremental/differential)

A cost or benefit is only relevant if it changes depending on which course of action is chosen. This is exactly why rent and utilities were excluded from the relevant-cost analysis in part (a): because they are identical whether the children's department operates or not, they have no bearing on the decision, even though they are genuine, ongoing cash costs of the business.

3. It must represent an actual cash flow

Relevant costs and benefits must correspond to real cash inflows or outflows arising from the decision. Purely accounting allocations that do not represent genuine incremental cash movements — such as an arbitrary share of head-office overhead allocated to a department regardless of whether it operates — are not relevant, even where they appear as a cost in a department's income statement.

Mark Allocation

Marks	Criteria
13–15	All three criteria correctly identified and clearly explained, using accurate terminology (future, incremental/differential, cash flow), with a specific link to the Baldwin Shoe Company scenario for at least one criterion.
9–12	All three criteria identified with reasonable explanation, or two explained in excellent depth.
5–8	One or two criteria identified with limited explanation.
0–4	No relevant criteria identified.

QUESTION 2 TOTAL — 50 MARKS

QUESTION 3**SmartWearables Ltd**

Total: 50 marks

SmartWearables Ltd produces the "Echo" smartwatch. The newly appointed Managing Director wants to assess 2023 trading performance, comparing budget to actual, where actual production and sales were 10% lower than the original 20,000-unit plan due to supply chain challenges. Candidates are required to (a) construct a flexible budget and calculate flexible budget variances and volume variances, (b) evaluate SmartWearables' financial performance based on that analysis, and (c) outline four disadvantages of budgeting.

Part	Requirement	Marks
(a)	Flexible budget, flexible budget variances (FBV) and volume variances (VV), each marked favourable or unfavourable	24
(b)	Evaluate 2023 financial performance and advise Mr. Smith	14
(c)	Four disadvantages of budgeting	12

Question 3(a)**24 marks**

SmartWearables Ltd, 2023, Echo smartwatch: Budget (based on 20,000 units) and Actual figures given for sales revenue, variable costs (direct materials, direct labour, production overheads, marketing, distribution) and fixed costs (insurance, salaries, rent, utilities, other). Actual production and sales were 10% lower than the 20,000-unit plan. (a) Construct a flexible budget for 2023 in relation to Echo. Then, using the flexible budget, master budget and actual amounts, calculate flexible budget variances (FBVs) and volume variances (VVs) for each budgetary item, indicating whether each variance should be considered favourable (F) or unfavourable (U).

Full Model Answer**Working - actual volume and per-unit budget rates**

Item	Value
Master budget volume	20,000 units
Actual volume (10% lower than plan)	18,000 units
Budgeted selling price per unit ($\text{€}2,000,000 \div 20,000$)	€100.00

Method: The flexible budget re-states the master budget's variable items at the *actual* volume of 18,000 units, using the *budgeted* rate per unit for each; fixed costs remain unchanged in the flexible budget. The flexible budget variance (FBV) isolates genuine spending/price/efficiency differences at the same, actual volume; the volume variance (VV) isolates the effect of the volume shortfall alone, at budgeted rates.

Full Model Answer (continued) - Flexible budget and variances: variable items

Flexible budget, FBV and VV - sales revenue and variable costs

Item	Master budget (€)	Flexible budget (€)	Actual (€)	FBV (Actual – Flex)	VV (Flex – Master)
Sales revenue	2,000,000	1,800,000	1,750,000	(50,000) U	(200,000) U
Direct materials	600,000	540,000	550,000	(10,000) U	(60,000) F
Direct labour	450,000	405,000	390,000	15,000 F	(45,000) F
Production overheads	200,000	180,000	185,000	(5,000) U	(20,000) F
Marketing	90,000	81,000	84,000	(3,000) U	(9,000) F
Distribution	72,000	64,800	63,000	1,800 F	(7,200) F

Sign convention for costs vs revenue: For sales revenue, a shortfall against the comparison figure is unfavourable. For costs, a **lower** actual/flexible figure is favourable (less was spent than expected), even though the arithmetic subtraction (Actual – Flex, or Flex – Master) may be negative — each variance above is labelled F or U according to this economic meaning, not the raw arithmetic sign.

Contribution

Item	Master budget (€)	Flexible budget (€)	Actual (€)	FBV	VV
Contribution	588,000	529,200	478,000	(51,200) U	(58,800) U

Full Model Answer (continued) – Fixed costs and net profit

Flexible budget, FBV and VV – fixed costs

Item	Master budget (€)	Flexible budget (€)	Actual (€)	FBV (Actual – Flex)	VV (Flex – Master)
Insurance	50,000	50,000	70,000	(20,000) U	0
Salaries	150,000	150,000	180,000	(30,000) U	0
Rent	30,000	30,000	27,000	3,000 F	0
Utilities	22,500	22,500	26,250	(3,750) U	0
Other fixed costs	7,500	7,500	9,000	(1,500) U	0
Total fixed costs	260,000	260,000	312,250	(52,250) U	0

Why fixed costs have no volume variance: Fixed costs are, by definition, unaffected by the volume of activity within the relevant range, so the flexible budget figure for every fixed cost line equals the master budget figure — the volume variance for every fixed cost item is therefore zero, and any difference between budget and actual is entirely a flexible budget (spending) variance.

Net profit

Item	Master budget (€)	Flexible budget (€)	Actual (€)	FBV	VV
Net profit	328,000	269,200	165,750	(103,450) U	(58,800) U

Overall reconciliation: The total variance in net profit (Actual €165,750 – Master budget €328,000 = €(162,250) unfavourable) is fully explained by the sum of the flexible budget variance (€(103,450) unfavourable) and the volume variance (€(58,800) unfavourable): €(103,450) + €(58,800) = €(162,250).

Mark Allocation

Marks	Criteria
20–24	Actual volume correctly derived (18,000 units); flexible budget correctly constructed for all variable items at budgeted per-unit rates; fixed costs correctly held constant; all FBVs and VVs correctly calculated and correctly labelled F/U; net profit reconciliation correct.
14–19	Substantially correct with minor errors in one or two line items, or F/U labelling inconsistently applied.
7–13	Flexible budget concept understood but significant errors in the FBV/VV split, or fixed costs incorrectly flexed with volume.
0–6	Limited understanding of flexible budgeting; minimal correct workings.

Question 3(b)**14 marks**

Based upon your answer to (a), briefly evaluate SmartWearables Ltd. financial performance during 2023 in relation to "Echo", offering whatever advice to Mr. Smith that you consider appropriate.

Full Model Answer**Headline result**

Actual net profit of €165,750 fell €162,250 (49.5%) short of the original €328,000 budget — a very substantial shortfall. Comparing actual straight to the master budget would blend two distinct issues together; the variance analysis in part (a) separates them so each can be assessed on its own merits.

The volume variance (€(58,800) unfavourable) reflects the supply chain disruption

This portion of the shortfall stems directly from the 10% fall in production and sales volume, described as an unexpected short-term supply chain challenge. While this raises questions about SmartWearables' supply chain resilience, it does not, on its own, reflect poor day-to-day operational management during the period.

The flexible budget variance (€(103,450) unfavourable) is the larger, more concerning component

This is nearly double the volume variance and represents genuine spending and efficiency problems *after* already adjusting for the lower volume achieved. Within this: direct labour and distribution costs were favourable (better cost control at the actual volume), but direct materials, production overheads and marketing were all unfavourable, and — most significantly — fixed costs overspent by €52,250 against budget, driven overwhelmingly by **salaries** (€(30,000) unfavourable) and **insurance** (€(20,000) unfavourable). Together these two items account for the great majority of the entire flexible budget variance and merit urgent, specific investigation.

Advice to Mr. Smith

Prioritise investigating the salary and insurance overruns, as these are by far the largest controllable drivers of the shortfall — possible causes include unbudgeted headcount growth, pay increases, or an insurance premium increase that was not reflected in the original budget. Recognise that variable cost control was mixed but not catastrophic once volume is properly accounted for. Separately, given that the volume shortfall alone accounts for over a third of the total profit variance, SmartWearables should review the resilience of its supply chain arrangements to reduce the risk of a repeat disruption affecting future periods.

Mark Allocation

Marks	Criteria
12-14	Correctly separates the volume-driven shortfall (largely uncontrollable) from the flexible budget variance (largely controllable); identifies salaries and insurance as the dominant drivers of the unfavourable FBV; gives specific, evidence-based advice referencing the figures from part (a).
8-11	Good evaluation distinguishing FBV from VV with reasonable identification of key drivers; advice more generic.
4-7	Some relevant commentary but does not clearly distinguish the two variance types, or lacks specific reference to individual line items.
0-3	Minimal or no meaningful evaluation.

Question 3(c)**12 marks**

Outline and briefly explain FOUR disadvantages associated with the use of budgeting that Mr. Smith should be aware of.

Full Model Answer**1. Time-consuming and costly to prepare**

Preparing a detailed annual budget can consume significant management time and organisational resources that could otherwise be spent on value-adding activities. This cost must be weighed against the benefits the budgeting process provides.

2. A fixed master budget can quickly become an inappropriate basis for comparison

Once actual conditions diverge from the plan — as with SmartWearables' 10% volume shortfall — comparing actual results directly against the original master budget produces a misleading picture, as shown by the very different (and much less alarming) flexible budget variance versus the raw budget-to-actual comparison. A flexible budgeting adjustment, as performed in part (a), is required to give a fair, like-for-like assessment, adding further complexity to the process.

3. Can encourage dysfunctional behaviour and budgetary slack

Where managers are evaluated against meeting budget targets, this can create an incentive to deliberately understate achievable revenue or overstate necessary costs when the budget is set (building in slack to make future targets easier to hit), or to spend up to the full remaining budget near the end of a period purely to avoid a reduced allocation next year, regardless of genuine need.

4. Can encourage a short-term focus at the expense of longer-term strategy

Because managers are often evaluated primarily against whether they meet the current year's budget, budgeting can bias decision-making towards actions that improve short-term reported results (e.g. deferring necessary maintenance or investment spend) at the expense of the organisation's longer-term strategic health.

Other acceptable disadvantages: Candidates may alternatively discuss: budgets can quickly become outdated in a fast-changing or uncertain trading environment (as illustrated by the unexpected supply chain disruption in this question); an incremental approach can simply carry forward existing inefficiencies from period to period without genuine reassessment; or budgeting can create unhelpful silo behaviour between departments competing for a limited overall allocation. Any four well-explained, distinct disadvantages receive full credit.

Mark Allocation

Marks	Criteria
10-12	Four distinct, well-explained disadvantages given, each a genuine and recognised limitation of budgeting; ideally at least one linked to the specific issues encountered in this question (e.g. the need for flexible budgeting given the volume shortfall).
7-9	Four disadvantages identified with reasonable explanation, or three explained in excellent depth.
4-6	Two to three disadvantages identified with limited explanation.
0-3	One or no relevant disadvantages identified.

QUESTION 3 TOTAL — 50 MARKS